

One Poisson Equation for Conformal Coverage under Dependence: Exact State Correction and the Honesty Margin

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Abstract

Conformal prediction under dependent calibration data has two coupled defects. Dependence deforms the law of the order-statistic threshold, and the current dynamical state shifts the law of the future score. We show a single Poisson equation controls both. The realized conditional coverage of the empirical-quantile threshold decomposes exactly into its invariant-law value plus a current-state corrector, with no asymptotics. The same Poisson operator supplies the long-run calibration variance, so state bias and effective-sample-size loss are two faces of one fast eigenmode. A Freedman bound then yields a finite-sample, state-conditionally honest quantile correction.

1 The exact object

Take a stationary time series of nonconformity scores and the ordinary split-conformal threshold, the empirical $(1 - \alpha)$ -quantile of a calibration block. Two things are true at once and are usually conflated. The calibration scores are dependent, so the order statistic used as the threshold is not the one an i.i.d. analysis assumes. And the test point is generated by a dynamical process whose current state is not the invariant state, so the law of the score being covered is not the invariant law. The exact object of study is neither the invariant law nor the marginal coverage. It is the realized conditional coverage of a random order-statistic threshold, and both effects act on it.

It is tempting to say that ordinary conformal prediction “calibrates the invariant law.” Under dependence that is not an exact guarantee. Dependence alters the law of the order statistic, and the current state alters the law of the future test score. The contribution here is that a single Poisson equation governs both perturbations at once: the transient gap between the current-state and invariant score laws, and the long-run calibration noise of the dependent empirical quantile. The central identity, equation (4) below, is exact and asymptotics-free, and we verify it numerically to machine precision.

Related work. Two effects of this kind have been described before, and we treat them as background rather than as the contribution. Transported-beta analyses separate a test-side coverage transport from a deformation of the calibration order statistic, with the Gaussian AR(1) case worked out explicitly. Markov distributional conformal prediction constructs a conditional probability integral transform from estimated transition CDFs. What the Poisson view adds is operational: the same operator that supplies the state corrector also supplies a martingale

representation of the calibration term and its long-run variance, which is what the honesty margin needs. The nonparametric price of a fixed-point guarantee is the subject of a companion note (Cotton, 2026), whose local modulus reappears in Section 9.

2 The exact coverage decomposition

Let (W_t) be a Markov chain with transition kernel P and invariant distribution π , on a state space rich enough to absorb any emission randomness, so that the score is a deterministic readout $R_t = r(W_t)$. (Replacing the indicator below by a conditional emission CDF costs nothing.) For each threshold q write the coverage function and the invariant score CDF

$$H_q(w) = \mathbf{1}\{r(w) \leq q\}, \quad F(q) = \pi H_q. \quad (1)$$

Let $R_{(k)}$ be the k th order statistic of the calibration scores $R_1, \dots, R_n$, and let $\mathcal{F}_n$ be the calibration information. If the test score is observed ℓ steps after the calibration block, its realized conditional coverage is

$$C_{n,\ell,k} = \mathbb{P}\{R_{n+\ell} \leq R_{(k)} \mid \mathcal{F}_n\}. \quad (2)$$

Conditional on the present state W_n , the Markov property gives $C_{n,\ell,k} = P^\ell H_{R_{(k)}}(W_n)$, the ℓ -step transported coverage function evaluated at the current state.

The invariant law is the wrong reference by exactly a solved Poisson equation. Fix q and solve

$$(I - P)\chi_q = H_q - F(q), \quad \pi\chi_q = 0, \quad (3)$$

which has a unique centered solution under standard ergodicity (Meyn and Tweedie, 2009). Since $F(q)$ is constant and P^ℓ fixes constants,

$$P^\ell H_q - F(q) = P^\ell (I - P)\chi_q = (P^\ell - P^{\ell+1})\chi_q.$$

Theorem 1 (Exact conformal coverage corrector). *For every calibration threshold $R_{(k)}$,*

$$C_{n,\ell,k} = F(R_{(k)}) + (P^\ell - P^{\ell+1})\chi_{R_{(k)}}(W_n). \quad (4)$$

The first term is the coverage of the random calibration threshold under the invariant score law. The second is the exact current-state correction. If the state is latent with filtering distribution $\mu_n(\cdot) = \mathbb{P}(W_n \in \cdot \mid \mathcal{F}_n)$, then

$$C_{n,\ell,k} = F(R_{(k)}) + \mu_n(P^\ell - P^{\ell+1})\chi_{R_{(k)}}. \quad (5)$$

No asymptotics enter. The decomposition is an algebraic identity given the Poisson solution, and it holds for every n, ℓ, k . A direct simulation on a finite chain confirms both sides of (4) agree to machine precision.

3 A transported order-statistic representation

Suppose F is continuous and set $U_i = F(R_i)$. Each U_i is marginally uniform under stationarity, though the sequence need not be independent, and monotonicity of F gives $F(R_{(k)}) = U_{(k)}$. Define the state-dependent coverage transport

$$\tau_{\ell,w}(u) = P^\ell H_{F^{-1}(u)}(w) = u + (P^\ell - P^{\ell+1})\chi_{F^{-1}(u)}(w). \quad (6)$$

Then Theorem 1 reads

$$C_{n,\ell,k} = \tau_{\ell,W_n}(U_{(k)}), \quad (7)$$

which separates the problem cleanly. The law of $U_{(k)}$ carries the calibration dependence. The map τ_{ℓ,W_n} carries the test-side state dependence. Under i.i.d. calibration $U_{(k)}$ has the familiar beta law and τ is the identity, recovering ordinary conformal coverage. Under dependent calibration the law of $U_{(k)}$ is generally not beta, and its leading variance is set by the long-run variance of the coverage indicator, computed in Section 5. This is the transported-beta separation, to which the Poisson representation adds the explicit corrector and a martingale representation of the calibration term.

4 The first-order coverage expansion

Fix $p = 1 - \alpha$ and $q_p = F^{-1}(p)$, and abbreviate the transient corrector

$$b_{\ell,w}(q) = P^\ell H_q(w) - F(q) = (P^\ell - P^{\ell+1})\chi_q(w). \quad (8)$$

The dependent empirical quantile obeys a Bahadur representation

$$F(R_{(k_p)}) - p = -[F_n(q_p) - p] + o_p(n^{-1/2}), \quad F_n(q) = \frac{1}{n} \sum_{i=1}^n \mathbf{1}\{R_i \leq q\}, \quad (9)$$

valid under geometric ergodicity with F differentiable at q_p , $f(q_p) > 0$. Expanding (4) around q_p and using (9),

$$C_{n,\ell,k_p} - p = b_{\ell,W_n}(q_p) - \left[1 + \frac{1}{f(q_p)} \partial_q b_{\ell,W_n}(q_p)\right] [F_n(q_p) - p] + o_p(n^{-1/2}). \quad (10)$$

When the transient correction is small in C^1 , $\|b_{\ell,\cdot}\|_{C^1} = o(1)$, this reduces to the central formula

$$C_{n,\ell,k_p} - p = b_{\ell,W_n}(q_p) - [F_n(q_p) - p] + o_p(\|b_{\ell,\cdot}\| + n^{-1/2}). \quad (11)$$

Realized coverage error has two first-order sources: a current-state bias $b_{\ell,W_n}(q_p)$ and a calibration-CDF error $F_n(q_p) - p$. The first-order probability-level correction is therefore

$$p \mapsto p - b_{\ell,W_n}(q_p). \quad (12)$$

If the current state causes undercoverage then $b_{\ell,W_n}(q_p) < 0$ and the required empirical quantile level rises. A one-sided honesty margin is then added on top, and its size is set by the same operator, as we show next.

5 The same Poisson equation gives the calibration variance

Center the coverage indicator at the invariant quantile,

$$g_p(w) = H_{q_p}(w) - p, \quad (13)$$

and solve a second Poisson equation with the same operator,

$$(I - P)\psi_p = g_p, \quad \pi\psi_p = 0. \quad (14)$$

The additive calibration error has the exact martingale decomposition

$$\sum_{i=0}^{n-1} g_p(W_i) = \psi_p(W_0) - \psi_p(W_n) + \sum_{i=0}^{n-1} D_{i+1}, \quad D_{i+1} = \psi_p(W_{i+1}) - P\psi_p(W_i), \quad (15)$$

with (D_{i+1}) a martingale-difference sequence. Its long-run variance is

$$\sigma_p^2 = \pi[\psi_p^2 - (P\psi_p)^2] = p(1-p) + 2 \sum_{j=1}^{\infty} \text{Cov}_{\pi}(\mathbf{1}\{R_0 \leq q_p\}, \mathbf{1}\{R_j \leq q_p\}). \quad (16)$$

So one operator supplies both the transient bias $b_{\ell,w}(q) = (P^{\ell} - P^{\ell+1})\chi_q(w)$ and the calibration variance $\sigma_p^2 = \pi[\psi_p^2 - (P\psi_p)^2]$. These are not unrelated corrections. They are two manifestations of the same unresolved fast dynamics. The corrector $b_{\ell,w}$ measures how much fast-state information survives to the test horizon. The variance σ_p^2 measures how much fast-state persistence removes calibration information.

6 A fast two-state volatility model

The cleanest nontrivial instance is not AR(1) but a fast volatility regime. Let $V_t^{\varepsilon} \in \{L, H\}$ be a continuous-time chain with generator

$$Q_{\varepsilon} = \frac{1}{\varepsilon} \begin{pmatrix} -a & a \\ b & -b \end{pmatrix}, \quad \kappa = a + b, \quad \pi_H = \frac{a}{a+b}, \quad \pi_L = \frac{b}{a+b}. \quad (17)$$

Conditional on the state, the score CDFs are $F_L(q)$ and $F_H(q)$, and the invariant score law is the mixture $F(q) = \pi_L F_L(q) + \pi_H F_H(q)$. If the filtered high-state probability at time zero is $\omega_0 = \mathbb{P}(V_0 = H | \mathcal{F}_0)$, then at horizon t

$$\omega_t = \pi_H + e^{-\kappa t/\varepsilon}(\omega_0 - \pi_H), \quad (18)$$

so the exact conditional score CDF is

$$G_t^{\varepsilon}(q) = F(q) + e^{-\kappa t/\varepsilon}(\omega_0 - \pi_H)[F_H(q) - F_L(q)], \quad (19)$$

and the transient coverage error is

$$b_t^{\varepsilon}(q) = e^{-\kappa t/\varepsilon}(\omega_0 - \pi_H)[F_H(q) - F_L(q)]. \quad (20)$$

This is a literal homogenization boundary layer. After several fast mixing times it is exponentially small. At a forecast horizon comparable to ε it is not small at all. To first order the quantile correction is

$$q_{t,p}^{\varepsilon} = q_p - \frac{e^{-\kappa t/\varepsilon}(\omega_0 - \pi_H)[F_H(q_p) - F_L(q_p)]}{f(q_p)} + O(e^{-2\kappa t/\varepsilon}), \quad f = \pi_L f_L + \pi_H f_H. \quad (21)$$

For a nonnegative absolute-error score the high-volatility state usually has $F_H(q) < F_L(q)$ at the relevant q . If the current high-volatility probability exceeds stationary, $\omega_0 > \pi_H$, then $b_t^{\varepsilon}(q_p) < 0$ and (21) raises the threshold, as it should.

Discrete sampling and calibration dependence. Sample every Δ time units. The non-trivial eigenvalue of the sampled transition matrix is $\lambda = e^{-\kappa\Delta/\varepsilon}$. Writing $d_p = F_H(q_p) - F_L(q_p)$, the coverage indicator $I_j = \mathbf{1}\{R_j \leq q_p\}$ has autocovariance $\text{Cov}(I_0, I_j) = \pi_L \pi_H d_p^2 \lambda^j$ for $j \geq 1$, so

$$\sigma_p^2 = p(1-p) + 2\pi_L \pi_H d_p^2 \frac{\lambda}{1-\lambda}. \quad (22)$$

The same quantities control both problems,

$$\underbrace{b_m^\varepsilon(q_p) \propto \lambda^m (\omega_0 - \pi_H) d_p}_{\text{transient bias}}, \quad \underbrace{\sigma_p^2 - p(1-p) \propto \pi_L \pi_H d_p^2 \frac{\lambda}{1-\lambda}}_{\text{calibration variance inflation}}. \quad (23)$$

One fast eigenmode produces both conditional miscoverage and effective-sample-size loss. When $\Delta \ll \varepsilon$ we have $1 - \lambda \sim \kappa\Delta/\varepsilon$, so $\lambda/(1-\lambda) \sim \varepsilon/(\kappa\Delta)$. Sampling faster then piles up redundant calibration scores, and for a window of fixed physical duration $T = n\Delta$ the effective sample size saturates at order T/ε rather than growing like n .

A numerical illustration. Take $p = 0.9$, $\pi_H = 0.2$, $\lambda = 0.8$, and at the invariant 0.9-quantile $F_L(q_p) = 0.95$, $F_H(q_p) = 0.70$, so the mixture is $0.8(0.95) + 0.2(0.70) = 0.90$. If the current state is high volatility the one-step coverage of the invariant threshold is $0.9 + 0.8(1-0.2)(0.70-0.95) = 0.74$. If it is low volatility it is $0.9 + 0.8(0-0.2)(0.70-0.95) = 0.94$. The same pooled threshold gives either 74% or 94% conditional coverage while appearing perfectly calibrated at 90% in stationarity. The long-run variance is $\sigma_p^2 = 0.09 + 2(0.8)(0.2)(0.25)^2(0.8/0.2) = 0.17$, an integrated dependence factor $\sigma_p^2/[p(1-p)] = 0.17/0.09 \approx 1.89$, so the nominal calibration sample behaves like only $\approx 53\%$ as many independent observations. A direct simulation of the sampled chain reproduces σ_p^2 . The persistent regime creates severe current-state undercoverage and nearly halves the effective calibration sample at the same time.

7 A finite-sample honest Markov correction

The expansion motivates an actual finite-sample theorem. For horizon ℓ define the worst adverse transient discrepancy

$$a_\ell = \sup_{w,q} [F(q) - P^\ell H_q(w)]_+, \quad \text{so} \quad P^\ell H_q(w) \geq F(q) - a_\ell \quad (24)$$

for every state and threshold. Choose the invariant target level $u = p + a_\ell$, set $q_u = F^{-1}(u)$ and $g_u(w) = H_{q_u}(w) - u$, and solve $(I - P)\psi_u = g_u$. Assume the uniform bounds

$$\|\psi_u\|_\infty \leq B, \quad P\psi_u^2(w) - (P\psi_u(w))^2 \leq V \quad \text{uniformly in } w. \quad (25)$$

With $x = \log(1/\eta)$ define the deviation

$$\delta_{n,\eta} = \frac{2B + \sqrt{2nVx}}{n} + \frac{2Bx}{3n}, \quad (26)$$

and set $k = \lfloor n(u + \delta_{n,\eta}) \rfloor + 1$, with $R_{(k)} = +\infty$ if $k > n$.

Theorem 2 (Finite-sample honest Markov correction). *Under the preceding assumptions,*

$$\mathbb{P}\left\{\mathbb{P}(R_{n+\ell} \leq R_{(k)} \mid \mathcal{F}_n) \geq p\right\} \geq 1 - \eta. \quad (27)$$

Proof. By the martingale decomposition (15) applied at q_u , $\sum_{i=1}^n [\mathbf{1}\{R_i \leq q_u\} - u]$ equals a martingale plus endpoint terms bounded by $2B$. Freedman's inequality (Freedman, 1975) with the variance bound (25) gives, with probability at least $1 - \eta$,

$$F_n(q_u) \leq u + \delta_{n,\eta}.$$

On this event fewer than k calibration scores lie at or below q_u , since $nF_n(q_u) \leq n(u + \delta_{n,\eta}) < k$, so $R_{(k)} \geq q_u$ and hence $F(R_{(k)}) \geq u = p + a_\ell$. By the definition of a_ℓ ,

$$\mathbb{P}(R_{n+\ell} \leq R_{(k)} | \mathcal{F}_n) = P^\ell H_{R_{(k)}}(W_n) \geq F(R_{(k)}) - a_\ell \geq p,$$

which is (27). $\square$

This is a genuine calibration-sample honest and state-conditionally valid guarantee, stronger than a statement about average marginal coverage. If B , V , and a_ℓ hold uniformly over a dynamic model class, the guarantee is uniform over that class.

8 Interpretation

The corrected empirical quantile level is

$$p + \underbrace{a_\ell}_{\text{fast-state protection}} + \underbrace{\delta_{n,\eta}}_{\text{dependent-calibration protection}}, \quad (28)$$

and in threshold units the first-order inflation is

$$T_n - q_p \approx \frac{a_\ell + \delta_{n,\eta}}{f(q_p)}. \quad (29)$$

The two costs are qualitatively different. The transient term a_ℓ is not an estimation error. It is the real difference between the current conditional law and the invariant law. The term $\delta_{n,\eta}$ is the price of demanding that the realized calibration sample be trustworthy with probability $1 - \eta$. A state-adaptive implementation need not pay the worst-case a_ℓ . It can estimate $b_{\ell, W_n}(q) = (P^\ell - P^{\ell+1})\chi_q(W_n)$ and build a lower confidence envelope for the current conditional CDF. Theorem 2 is the simplest version with a transparent proof.

9 Combining with the local sharpness modulus

The companion note (Cotton, 2026) asks for a stronger guarantee: with high probability over calibration data, the realized threshold must cover the conditional score law at a fixed local covariate value x_0 . It proves that this forces inflation at the local nonparametric modulus $(ng(x_0))^{-s/(2s+d)}$, from locally indistinguishable alternatives rather than any exchangeability phenomenon. With dependent fast-slow scores the natural upper bound becomes, up to fixed regularity constants,

$$T_{n,h}(x_0) - q_p(x_0) \lesssim C_q L h^s + \frac{\sigma_{p,x_0} \sqrt{\log(1/\eta)}}{f_{p,x_0} \sqrt{ng(x_0)h^d}} + \frac{a_{\ell,x_0}}{f_{p,x_0}} + R_{\text{hom}}(\varepsilon) + \Delta_{\text{dyn}}. \quad (30)$$

The terms are local smoothing bias $C_q L h^s$, dependent local calibration noise through the long-run variance σ_{p,x_0} , remaining fast-state memory a_{ℓ,x_0} , the homogenization approximation error

$R_{\text{hom}}(\varepsilon)$, and the cost Δ_{dyn} of estimating the transition dynamics, filter, and cell problem. Minimizing the first two terms over h gives

$$h_{\star} \asymp \left[\frac{\sigma_{p,x_0}^2}{ng(x_0)L^2} \right]^{1/(2s+d)}, \quad \text{local inflation} \asymp L^{d/(2s+d)} \left[\frac{\sigma_{p,x_0}^2}{ng(x_0)} \right]^{s/(2s+d)}. \quad (31)$$

Dependence changes the information constant through the long-run variance but leaves the Hölder exponent unchanged. Homogenization adds the transient and dependence terms and does not erase the local modulus. The result is a clean hierarchy,

$$\begin{aligned} \text{local heterogeneity} &\rightarrow \text{nonparametric modulus,} \\ \text{calibration persistence} &\rightarrow \text{Poisson long-run variance,} \\ \text{current fast state} &\rightarrow \text{Poisson transient corrector,} \\ \text{approximating the fast-slow model} &\rightarrow \text{homogenization remainder.} \end{aligned} \quad (32)$$

The first term is unavoidable over a broad Hölder class. The second and third can be quantified and corrected under dynamical structure. The fourth must be included for the guarantee to be honest.

10 What homogenization does and does not buy

This settles the conceptual question. Homogenization does not defeat the sharpness lower bound. It partitions the conformal error into unavoidable local ambiguity, correctable dynamical bias, and estimable dependence noise. If the fast-slow dynamics live in a finite-dimensional parametric family, honest uncertainty about the dynamical correction shrinks at the parametric rate $n^{-1/2}$ rather than the nonparametric local rate $n^{-s/(2s+d)}$. That gain does not contradict the companion note. It comes from replacing a broad Hölder class by a much smaller structured experiment. If the transition law, the state-specific score distributions, or the scale field are again allowed to vary nonparametrically with x , the local modulus returns, and if their smoothness is unknown the no-adaptation obstruction returns with it.

11 The corresponding skaters transform

The correction is directly usable. For an existing distributional forecast F_{t+1}^0 , compute the Gaussianized probability integral transform $Z_t = \Phi^{-1}(F_t^0(Y_t))$, and fit a predictable two-state model to the past Z_t with state-specific CDFs K_L, K_H and filtered high-state probability ω_t . The one-step forecast is $\omega_{t+1|t} = \pi_H + \lambda(\omega_t - \pi_H)$, so the conditional CDF of the next Gaussianized PIT is

$$K_{t+1|t}(z) = (1 - \omega_{t+1|t})K_L(z) + \omega_{t+1|t}K_H(z), \quad (33)$$

and the dynamically corrected predictive CDF and u -quantile are

$$F_{t+1}(y) = K_{t+1|t}[\Phi^{-1}(F_{t+1}^0(y))], \quad \hat{Q}_{t+1}(u) = (F_{t+1}^0)^{-1}[\Phi(K_{t+1|t}^{-1}(u))]. \quad (34)$$

This is a direct replacement for the marginal fence-post transform. The fence-post map estimates a pooled PIT law. Equation (34) estimates the conditional PIT law after accounting for the current residual regime. For an honest upper quantile, let $\mathcal{C}_t$ be a confidence set for the dynamic parameters and add a homogenization or misspecification radius r_t ,

$$z_{t+1,p}^{\text{hon}} = \sup_{\theta \in \mathcal{C}_t} K_{\theta,t+1|t}^{-1}(p + r_t), \quad y_{t+1,p}^{\text{hon}} = (F_{t+1}^0)^{-1}[\Phi(z_{t+1,p}^{\text{hon}})]. \quad (35)$$

Empirical protocol. A comparison should include the base forecast, the marginal fence-post correction, an AR correction, the two-state homogenized correction, and the honest two-state correction. The decisive diagnostics are PIT calibration conditional on the filtered state, coverage conditional on recent Z_t and volatility, the lower-tail probability of realized coverage, proper scores such as log score and CRPS, and the extra interval width the honest envelope demands. Aggregate PIT uniformity alone is insufficient. The two-state example shows that pooled 90% calibration can conceal conditional coverage ranging from 74% to 94%.

12 Conclusion

The conformal coverage process under Markov dependence admits an exact invariant-plus-Poisson decomposition, equation (4), and the same Poisson equation determines both the current-state coverage bias and the dependence-adjusted honesty margin. State bias and effective-sample-size loss are two readings of one fast eigenmode. Placed next to the local sharpness modulus, this identifies precisely what dynamical structure can and cannot buy back: it corrects the transient and dependence terms, it does not remove the nonparametric floor, and honest use requires paying for the approximation of the fast–slow model itself.

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